Recast Error Intervals, Refinement Size, and Recall Bounds
A Complete Technical Report

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Abstract

This report develops the complete theoretical chain in Recast, from quantization error to refinement size and a lower bound on refinement-stage Recall. The analysis has three layers. First, an exact error identity and a static refinement certificate require no probabilistic assumptions. Second, a high-dimensional isotropic approximation yields candidate-dependent error scales and probabilistic distance intervals. Third, independent Gaussian errors, approximately equal error scales, and a local distance-density model yield interpretable estimates of the expected refinement size and expected Recall. The report also resolves the dependence created by the lower-bound error shared across pairwise retention events: conditional independence and positive association of comonotone functions make the product of marginal probabilities a valid lower bound under the stated assumptions. For $K = 10$, $z_p = 2$, $\rho = 10$, and $\rho' = 10000$, the model predicts about 50 refined candidates and an expected refinement-stage Recall lower bound of approximately 0.990243.

1 Problem Setting and Notation

Let $q \in \mathbb{R}^D$ be a query and let $\mathcal{C}(q) = \{x_1, \ldots, x_n\}$ be the candidate pool produced by the coarse search stage. We use squared Euclidean distance:

$$d(q, x) = \|q - x\|_2^2.$$

Order the candidates by their exact distances, using a fixed deterministic tie-breaking rule:

$$d_1 \le d_2 \le \cdots \le d_n, \qquad d_i = d(q, x_i).$$

The goal is to determine which candidates must be refined without reading all original vectors from external storage. The exact Top-K set within the candidate pool is

$$\mathrm{TopK}_{\mathcal{C}}(q) = \{x_1, \ldots, x_K\}.$$

For each candidate x_i, the in-memory code provides an estimated distance $\hat{d}_i$, an error scale τ_i, and a candidate distance interval with radius $z_p \tau_i$:

$$L_i = \hat{d}_i - z_p \tau_i, \qquad U_i = \hat{d}_i + z_p \tau_i.$$

Let Φ and ϕ denote the CDF and PDF of the standard normal distribution, respectively, and define

$$z_p = \Phi^{-1}\left(\frac{1+p}{2}\right).$$

Then $p = 2\Phi(z_p) - 1$ is the two-sided marginal coverage probability for a single candidate under the Gaussian model.

Two kinds of statements are kept separate throughout this report:

- **Deterministic statement.** If every interval used by the certificate is valid, the static certificate cannot omit any exact Top-K candidate within the candidate pool.
- **Model-based statement.** The numerical conclusions of roughly 50 refinements and expected Recall above 0.99 depend on Gaussian errors, independence across candidates, and a local distance-spacing model.

2 Exact Error Decomposition for Norm-Aware Encoding

Let μ be the training center and define

$$q_c = q - \mu, \qquad z_x = x - \mu, \qquad a_x = \|z_x\|_2, \qquad u_x = \frac{z_x}{a_x}.$$

The encoder stores the norm a_x and encodes the unit direction u_x, which is reconstructed as $\tilde{u}_x$. Define the estimated distance as

$$\hat{d}(q, x) = \|q_c\|_2^2 + a_x^2 - 2a_x q_c^\top \tilde{u}_x.$$

Lemma 2.1 (Distance error identity). *Let $e_x = u_x - \tilde{u}_x$ be the directional reconstruction error. Then*

$$d(q, x) - \hat{d}(q, x) = -2a_x q_c^\top e_x.$$

Proof. Because $x - \mu = a_x u_x$,

$$d(q, x) = \|q_c - a_x u_x\|_2^2 = \|q_c\|_2^2 + a_x^2 - 2a_x q_c^\top u_x.$$

Subtracting $\hat{d}(q, x)$ gives

$$d(q, x) - \hat{d}(q, x) = -2a_x q_c^\top (u_x - \tilde{u}_x) = -2a_x q_c^\top e_x.$$

□

This identity immediately gives a deterministic error bound that does not depend on any distributional assumption.

Corollary 2.2 (Deterministic Cauchy–Schwarz radius). *For any q and x,*

$$|d(q, x) - \hat{d}(q, x)| \le 2a_x \|q_c\|_2 \|e_x\|_2.$$

Consequently,

$$\epsilon_x^{\det} = 2a_x \|q_c\|_2 \|e_x\|_2$$

is the radius of an always-valid distance interval.

This deterministic radius is usually wide because it assumes worst-case alignment. Recast's probabilistic radius exploits the fact that a random projection in high dimensions is typically smaller by a factor on the order of $1/\sqrt{D}$.

3 High-Dimensional Projection Model and Probabilistic Intervals

Define

$$T_x = \frac{q_c^\top e_x}{\|q_c\|_2 \|e_x\|_2},$$

the cosine between the query direction and the encoding-error direction.

Proposition 3.1 (High-dimensional spherical projection)**.** *Suppose the normalized query direction is uniformly distributed on $\mathbb{S}^{D-1}$ relative to a fixed error direction e_x. Then*

$$\mathbb{E}[T_x] = 0, \qquad \mathrm{Var}(T_x) = \frac{1}{D},$$

and its density is proportional to $(1-t^2)^{(D-3)/2}$. Moreover, as $D \to \infty$,

$$\sqrt{D}\, T_x \xrightarrow{d} \mathcal{N}(0,1).$$

Under this high-dimensional isotropic approximation, conditional on a_x, $\|q_c\|_2$, and $\|e_x\|_2$,

$$d(q,x) - \hat{d}(q,x) \approx \mathcal{N}(0, \tau_x^2),$$

where the natural candidate-dependent error scale is

$$\boxed{\tau_x = \frac{2a_x \|q_c\|_2 \|e_x\|_2}{\sqrt{D}}}.$$

Proposition 3.2 (Single-candidate probabilistic interval)**.** *Under the Gaussian approximation, define*

$$L_x = \hat{d}(q,x) - z_p \tau_x, \qquad U_x = \hat{d}(q,x) + z_p \tau_x.$$

Then

$$\mathbb{P}\big(d(q,x) \in [L_x, U_x]\big) \approx p, \qquad p = 2\Phi(z_p) - 1.$$

Remark 3.3. The marginal coverage p of one candidate is not the probability that all intervals in the candidate pool are simultaneously valid. The certificate safety result in the next section is therefore a deterministic theorem conditioned on interval validity. Section 6 instead analyzes expected Recall directly under the probabilistic error model.

4 Static Distance Certificate

Let $U_{(K)}$ be the K-th smallest upper bound among all candidates and write

$$\theta_K = U_{(K)}.$$

The static Recast refinement set is

$$\mathcal{R}(q) = \{x_i \in \mathcal{C}(q) : L_i \le \theta_K\}.$$

Theorem 4.1 (Safety of the static certificate)**.** *If all candidate intervals are valid, i.e., $d_i \in [L_i, U_i]$, then*

$$\mathrm{TopK}_{\mathcal{C}}(q) \subseteq \mathcal{R}(q).$$

Proof. Consider any skipped candidate x_y, for which $L_y > \theta_K$. By the definition of θ_K, at least K candidates $x_{i_1}, \ldots, x_{i_K}$ satisfy $U_{i_r} \le \theta_K$. Interval validity implies

$$d_{i_r} \le U_{i_r} \le \theta_K < L_y \le d_y, \qquad r = 1, \ldots, K.$$

Thus at least K candidates have exact distances strictly smaller than d_y, so x_y cannot belong to the exact Top-K within the candidate pool. Every exact Top-K candidate is therefore refined. □

Proposition 4.2 (Minimality with interval-only information)**.** *Suppose a policy observes only the candidate intervals $[L_i, U_i]$ and must be safe for every assignment of exact distances consistent with those intervals. Then no candidate satisfying $L_i \le U_{(K)}$ can be safely skipped. Hence $\mathcal{R}(q)$ is the smallest safe refinement set for an interval-only policy.*

Proof. Fix a candidate x_i with $L_i \le U_{(K)}$ and set $d_i = L_i$. Fewer than K other candidates can have $U_j < L_i$; otherwise the K-th smallest upper bound would also be smaller than L_i. Set every other distance to its upper endpoint $d_j = U_j$. Under this interval-consistent assignment, at most $K-1$ candidates are forced to precede x_i, so x_i can belong to the Top-K. Any policy that must be safe for every consistent assignment must therefore retain every candidate satisfying $L_i \le U_{(K)}$. □

5 A Model of Refinement Size

For every candidate,

$$L_i = U_i - 2z_p\tau_i.$$

Therefore,

$$|\mathcal{R}(q)| = \#\{i : U_i - 2z_p\tau_i \le U_{(K)}\}.$$

The K candidates whose upper bounds do not exceed $U_{(K)}$ are necessarily included. Additional work comes from candidates to the right of $U_{(K)}$ whose intervals still overlap the boundary.

Assumption 5.1 (Locally uniform upper bounds and equal radii)**.** In the relevant neighborhood to the right of $U_{(K)}$, candidate upper bounds are approximately uniformly spaced by Δ_U, and their error scales are approximately equal, $\tau_i \approx \tau$.

Proposition 5.2 (Refinement size under the local-spacing model)**.** *Ignoring truncation at the end of the candidate pool,*

$$|\mathcal{R}(q)| = K + \left\lfloor \frac{2z_p\tau}{\Delta_U} \right\rfloor.$$

Allowing the error scale and local density to vary across queries gives the first-order approximation

$$\mathbb{E}[|\mathcal{R}(q)|] \approx K + \frac{2z_p\mathbb{E}[\tau]}{\Delta_U}.$$

Define the boundary-density ratio

$$\rho = \frac{\mathbb{E}[\tau]}{\Delta_U}.$$

Then

$$\boxed{\mathbb{E}[|\mathcal{R}(q)|] \approx K + 2z_p\rho}.$$

For $K = 10$, $z_p = 2$, and $\rho = 10$,

$$\mathbb{E}[|\mathcal{R}(q)|] \approx 10 + 2 \times 2 \times 10 = 50.$$

Remark 5.3 (Refined candidates versus I/O operations)**.** The value 50 describes the expected order of the number of refined candidates. It is not an unconditional guarantee of 50 random I/O operations. The page-read count is of the same order only when candidates reside on distinct pages. If multiple unresolved candidates share a page, page closure can reduce the number of page reads.

6 A General Lower Bound on Expected Recall

This section isolates Recall loss caused by the static refinement policy after the candidate pool has already been generated. Define the estimation error

$$\xi_i = \hat{d}_i - d_i.$$

Assumption 6.1 (Candidate error model)**.** Conditional on exact distances and error scales, candidate errors are mutually independent and satisfy

$$\xi_i \sim \mathcal{N}(0, \tau_i^2).$$

It follows that

$$L_i = d_i + \xi_i - z_p \tau_i, \qquad U_i = d_i + \xi_i + z_p \tau_i.$$

6.1 A Sufficient Event for Retaining a True Neighbor

Fix a true neighbor x_j with $j \le K$. For every non-neighbor $l > K$, define

$$A_{jl} = \{U_l > L_j\}.$$

Lemma 6.2 (Sufficient retention event)**.** *If $\bigcap_{l>K} A_{jl}$ occurs, then $x_j \in \mathcal{R}(q)$.*

Proof. If every $l > K$ satisfies $U_l > L_j$, only the other $K-1$ exact Top-K candidates can have upper bounds no greater than L_j. Thus at most $K-1$ candidates satisfy $U_i \le L_j$, which implies $U_{(K)} \ge L_j$. Therefore $L_j \le U_{(K)}$ and x_j is refined. Equality has probability zero in the continuous Gaussian model; a discrete implementation can use fixed tie-breaking. □

The probability that two candidates cross in the wrong direction is

$$\pi_{jl} \equiv \mathbb{P}(U_l \le L_j) \tag{1}$$

$$= \Phi\left(-\frac{(d_l - d_j) + z_p(\tau_l + \tau_j)}{\sqrt{\tau_l^2 + \tau_j^2}}\right). \tag{2}$$

Hence

$$\mathbb{P}(A_{jl}) = 1 - \pi_{jl} = \Phi\left(\frac{(d_l - d_j) + z_p(\tau_l + \tau_j)}{\sqrt{\tau_l^2 + \tau_j^2}}\right).$$

6.2 A Product Lower Bound with a Shared Lower-Bound Error

The events A_{jl} share the random variable ξ_j, so they are not unconditionally independent. Nevertheless, under the independent Gaussian error model, the product of their marginal probabilities remains a valid lower bound.

Lemma 6.3 (Positive association of comonotone functions)**.** *Let X be a scalar random variable, and let $f_1, \ldots, f_m$ be nonnegative functions that are all nonincreasing. Then*

$$\mathbb{E}\left[\prod_{r=1}^{m} f_r(X)\right] \ge \prod_{r=1}^{m} \mathbb{E}[f_r(X)].$$

Proof. For two functions with the same monotonicity, Chebyshev's integral inequality gives $\mathrm{Cov}(f(X), g(X)) \ge 0$. The product of nonnegative nonincreasing functions is still nonincreasing, so induction on the number of functions proves the result. □

Conditioned on $\xi_j = t$, the events A_{jl} depend on mutually independent variables ξ_l and are conditionally independent. Moreover,

$$p_{jl}(t) \equiv \mathbb{P}(A_{jl} \mid \xi_j = t) = \Phi\left(\frac{(d_l - d_j) + z_p(\tau_l + \tau_j) - t}{\tau_l}\right)$$

is nonincreasing in t. Therefore,

$$\mathbb{P}\left(\bigcap_{l>K} A_{jl}\right) = \mathbb{E}_{\xi_j}\left[\prod_{l>K} p_{jl}(\xi_j)\right] \tag{3}$$

$$\geq \prod_{l>K} \mathbb{E}_{\xi_j}[p_{jl}(\xi_j)] \tag{4}$$

$$= \prod_{l>K} \mathbb{P}(A_{jl}). \tag{5}$$

Theorem 6.4 (General candidate-dependent Recall lower bound)**.** *Under the candidate error model, for each $j \leq K$,*

$$\mathbb{P}(x_j \in \mathcal{R}(q)) \geq \prod_{l>K} \Phi\left(\frac{(d_l - d_j) + z_p(\tau_l + \tau_j)}{\sqrt{\tau_l^2 + \tau_j^2}}\right).$$

Consequently,

$$\boxed{\mathbb{E}[\text{Recall}@K] \geq \frac{1}{K}\sum_{j=1}^{K}\prod_{l>K} \Phi\left(\frac{(d_l - d_j) + z_p(\tau_l + \tau_j)}{\sqrt{\tau_l^2 + \tau_j^2}}\right)}.$$

7 Equal Error Scales and a Piecewise Distance-Density Model

To obtain an interpretable closed form, assume equal error scales.

Assumption 7.1 (Equal error scales)**.** For the candidates relevant to the current query, $\tau_i = \tau$.

Because $d_j \leq d_K$, the general Recall bound gives

$$\mathbb{P}(x_j \in \mathcal{R}(q)) \geq \prod_{l>K} \Phi\left(\frac{d_l - d_j}{\sqrt{2}\tau} + \sqrt{2}z_p\right) \tag{6}$$

$$\geq \prod_{h=1}^{n-K} \Phi\left(\frac{d_{K+h} - d_K}{\sqrt{2}\tau} + \sqrt{2}z_p\right). \tag{7}$$

The right-hand side is independent of j, and hence

$$\boxed{\mathbb{E}[\text{Recall}@K] \geq \prod_{h=1}^{n-K} \Phi\left(\frac{d_{K+h} - d_K}{\sqrt{2}\tau} + \sqrt{2}z_p\right)}. \tag{8}$$

Assumption 7.2 (Piecewise uniform spacing near the Top-K boundary)**.** Let

$$\rho = \frac{\tau}{\Delta}, \qquad \rho' = \frac{\tau}{\Delta'}, \qquad H = 2z_p\rho,$$

and temporarily assume that H is an integer. The first H candidates to the right of the boundary are spaced by Δ:

$$d_{K+h} - d_K = h\Delta = \frac{h\tau}{\rho}, \qquad 1 \leq h \leq H.$$

More distant candidates, starting beyond $2z_p\tau$, are spaced by Δ':

$$d_{K+H+t} - d_K = 2z_p\tau + t\Delta' = 2z_p\tau + \frac{t\tau}{\rho'}, \qquad t \geq 1.$$

Extending the tail of a finite candidate pool to infinitely many candidates can only decrease the product. Equation (8) therefore gives

$$\mathbb{E}[\text{Recall}\,@K] \geq \prod_{h=1}^{H} \Phi\Big(\frac{h}{\sqrt{2}\rho} + \sqrt{2}z_p\Big) \tag{9}$$

$$\times \prod_{t=1}^{\infty} \Phi\Big(\frac{t}{\sqrt{2}\rho'} + 2\sqrt{2}z_p\Big). \tag{10}$$

8 Integral Lower Bound and Numerical Result

Define

$$I(a) = \int_a^{\infty} \ln \Phi(x)\,\mathrm{d}x < 0.$$

The function $\ln \Phi(x)$ is increasing. For any increasing function f, a right-endpoint Riemann sum satisfies

$$\sum_{h=1}^{m} f(h/c) \geq \int_0^m f(t/c)\,\mathrm{d}t.$$

Taking logarithms in Equation (10) and applying this inequality to both products gives

$$\ln \mathbb{E}[\text{Recall}\,@K] \geq \sqrt{2}\rho \int_{\sqrt{2}z_p}^{2\sqrt{2}z_p} \ln \Phi(x)\,\mathrm{d}x \tag{11}$$

$$+ \sqrt{2}\rho' \int_{2\sqrt{2}z_p}^{\infty} \ln \Phi(x)\,\mathrm{d}x \tag{12}$$

$$= \sqrt{2}\rho I(\sqrt{2}z_p) + \sqrt{2}(\rho' - \rho) I(2\sqrt{2}z_p). \tag{13}$$

Theorem 8.1 (Integral Recall lower bound under the piecewise model)**.** *Under independent Gaussian errors, equal error scales, and the piecewise uniform spacing model,*

$$\boxed{\mathbb{E}[\text{Recall}\,@K] \geq \exp\Big(\sqrt{2}\rho I(\sqrt{2}z_p) + \sqrt{2}(\rho' - \rho) I(2\sqrt{2}z_p)\Big)}.$$

8.1 Explicit Control of the Far Tail

Let $Q(x) = 1 - \Phi(x)$. For $x \geq 0$, where $Q(x) \leq 1/2$, $\ln(1 - Q(x)) \geq -2Q(x)$, and therefore

$$I(a) \geq -2 \int_a^{\infty} Q(x)\,\mathrm{d}x.$$

Using the identity

$$\int_a^{\infty} Q(x)\,\mathrm{d}x = \phi(a) - aQ(a)$$

and the lower Mills-ratio bound

$$Q(a) \geq \frac{a}{a^2 + 1}\phi(a),$$

we obtain

$$I(a) \geq -\frac{2\phi(a)}{a^2 + 1} \geq -\frac{2\phi(a)}{a^2}.$$

Setting $a = 2\sqrt{2}z_p$ yields

$$I(2\sqrt{2}z_p) \geq -\frac{1}{4\sqrt{2\pi}} \frac{e^{-4z_p^2}}{z_p^2}.$$

This explains why, even when ρ' is large, candidates far from the Top-K boundary affect the Recall lower bound only through a term rapidly suppressed by $e^{-4z_p^2}$.

8.2 Numerical Reproduction

Set

$$K = 10, \qquad z_p = 2, \qquad \rho = 10, \qquad \rho' = 10000.$$

The marginal Gaussian coverage of one candidate interval is

$$2\Phi(2) - 1 \approx 0.954500.$$

Numerical integration gives

$$I(2\sqrt{2}) \approx -6.9198630039 \times 10^{-4},$$
$$I(4\sqrt{2}) \approx -1.2884487894 \times 10^{-9}.$$

Therefore,

$$\ln \mathbb{E}[\mathrm{Recall}\,@10] \geq \sqrt{2} \times 10 \times I(2\sqrt{2}) + \sqrt{2} \times 9990 \times I(4\sqrt{2}) \tag{14}$$
$$\approx -0.0098043673, \tag{15}$$

which implies

$$\boxed{\mathbb{E}[\mathrm{Recall}\,@10] \geq e^{-0.0098043673} \approx 0.99024354}.$$

At the same time, the refinement-size model gives

$$\boxed{\mathbb{E}[|\mathcal{R}(q)|] \approx 10 + 2 \times 2 \times 10 = 50}.$$

Remark 8.2. Although $z_p = 2$ gives approximately 95.45% marginal coverage for one candidate interval, the model-based expected Top-10 Recall can exceed 99%. These statements are compatible: Recall depends on relative ordering and distance gaps between candidates, not on the simultaneous validity of every interval in the candidate pool.

9 Scope and Limitations

1. **Candidate-pool Recall and refinement Recall are distinct.** The bound in this report concerns the refinement policy after the candidate pool is fixed. If coarse routing or IVF has insufficient candidate-pool Recall, end-to-end Recall is additionally limited by candidate generation.

2. **The Gaussian model is a strong assumption.** High-dimensional spherical projections motivate the approximation, but real query directions, quantization-error directions, and data distributions may be correlated. Boundary-error distributions and interval coverage should be measured on held-out or test queries.

3. **Independence across candidate errors is an assumption.** Quantization errors for candidates under the same query may be correlated. The product lower bound follows from conditional independence and positive association when candidate errors are independent; it cannot be reused under arbitrary dependence.

4. **Equal error scales and equal spacing are interpretive models.** A deployed system should use candidate-dependent τ_i and analyze performance using either the general bound or empirical statistics.

5. **The value 50 is a refinement size, not a fixed I/O bound.** Page layout, candidates per page, dynamic refinement, and page closure determine the actual number of random reads.

6. **Safety and efficiency are separable.** The Cauchy–Schwarz radius is deterministic but wide; the Gaussian radius is narrow and practical. Certificate logic depends only on interval validity, whereas efficiency depends on interval width.

10 Summary of the Theoretical Chain

Under the assumptions explicitly stated in this report, the Recast argument can be summarized as

$$\begin{aligned} d(q,x) - \hat{d}(q,x) &= -2a_x q_c^\top (u_x - \tilde{u}_x), \\ \tau_x &= \frac{2a_x \|q_c\|_2 \|u_x - \tilde{u}_x\|_2}{\sqrt{D}}, \\ \mathcal{R}(q) &= \{x_i : L_i \le U_{(K)}\}, \\ \mathbb{E}[|\mathcal{R}(q)|] &\approx K + 2z_p\rho, \\ \mathbb{E}[\text{Recall}\,@K] &\ge \frac{1}{K}\sum_{j=1}^{K}\prod_{l>K}\Phi\left(\frac{(d_l - d_j) + z_p(\tau_l + \tau_j)}{\sqrt{\tau_l^2 + \tau_j^2}}\right). \end{aligned}$$

For the piecewise uniform-spacing model with $K = 10$, $z_p = 2$, $\rho = 10$, and $\rho' = 10000$, this chain predicts about 50 refined candidates and a model-based expected Recall@10 lower bound of at least 0.990243.